Unify Learn — Developer Report (updated 21 October 2026)

Covers the 1 October 2026 test bug list (14 confirmed: 7 critical P0, 7 important P1), everything fixed since, what is left, and what still needs live verification — and why.

Current version: **v1.26.13**. Every change below was verified with `tsc --noEmit` both sides plus a production `vite build` before commit. Per-version history for v1.12–v1.24 is in the git log (`feat:` / `fix:` commits carry the version); this report picks up in detail at the 20 October batch.

P0 fixes (critical — blocked student testing)

- **BUG-001 Contributor sees all courses (v1.8.1).** Server gates (`requireCourseAccess`, 403 for collaborators outside their teaching enrollments) on weeks, topics, notes and publish; Browse/Studio only list assigned courses; admin can assign courses per user.
- **BUG-002 Raw math text (v1.8.2).** One MathJax renderer app-wide (`lib/mathjax.ts` + `MathText`); KaTeX uninstalled; service worker caches MathJax + fonts offline.
- **BUG-003 Diagram placeholders (v1.8.3).** `diagrams` Storage bucket + policies, `lib/storage.ts` upload, numbered figure cards, Studio diagram tagging.
- **BUG-004 Generation fails on large input (v1.8.4/v1.8.5).** Auto-split into 20k-char parts, per-part retry, typed errors (`INPUT_TOO_LARGE`, `BAD_OUTPUT`…), async convert jobs (`202 + poll`) so long generations survive timeouts.
- **BUG-005 Quiz leaks answers (v1.8.5).** Submit-at-end EOQ: neutral picks, `Check score` only when all answered, verdicts/feedback/record on submit.
- **BUG-006 ME 352 not found (v1.8.6).** `course_aliases` table + boot seed, `GET /v1/courses/search` across code/title/alias, Explore debounced search spanning both semesters with badges.
- **BUG-007 Silent stacking publishes (v1.8.7).** Occupied week/topic returns 409 (`WEEK_OCCUPIED` / `TOPIC_OCCUPIED`); Studio offers Replace / Add as new version / Cancel. Nothing overwrites or duplicates silently.

P1 fixes (important — before the research test)

- **BUG-008 Old + new versions both listed (v1.8.7 batch).** Student lists and the author dashboard group by topic and show the latest version only (`older` count, history in reader).
- **BUG-009 No Lecture 1/2/3 structure (v1.9.0).** Weeks hold lectures with per-lecture topic numbering; reader switches via `?c=` with cross-lecture Back/Next; progress, versions, resume and publish are lecture-keyed (migration `0018`). Studio/NoteBuilder lecture pickers; AI prompt emits lecture groups. **Known deviation:** EOQ stays one week bank after the final lecture — per-lecture banks have no storage yet (decision pending).
- **BUG-010 Partial EOQ banks (v1.9.1).** A short bank triggers one automatic repair pass requesting a fresh 10-question bank (8 MCQ + 2 FITB); repair never fails the job.
- **BUG-011 Definitions/formulas blend in (v1.9.2).** Symbol cards render through MathJax (no raw `\alpha` overlapping titles), flex-clamped, wider card spacing.
- **BUG-012 Dashboard strip (v1.9.3).** 2-column course-card grid (title, lecturer, weeks, topics done), one-tap elective removal, lecturer names batched into the catalog. **Decision pending:** BUG-012 asks for no separate Explore page — Explore stays as the add-courses path until you confirm removal (it is the only discovery route).
- **BUG-013 Slow page changes (v1.9.3).** Idle-time warmup of enrolled week-lists kills the slow-blank feel. **Note:** no discussion panel exists in the app, so that half of the report had nothing to fix.
- **BUG-014 Quiz extra click + unstyled Check (v1.9.2).** `Check score` / `Try again` styled like primary buttons (the class had zero CSS); quiz tab opens directly.

Post-list fixes

- **Explore crash (v1.9.4).** Search `useEffect` sat below the loading return → hook-count crash on load. All 14 routes audited; Explore was the sole offender.
- **Share live-links 404 (v1.11.1).** `GET /share/mine` was registered after `GET /share/:token` , so Express swallowed "mine" as a token. Route order fixed; create, resolve, revoke, WhatsApp unfurl re-verified by code path.
- **My Classes + offline-first (v1.10.0).** Recurring weekly timetable + live roster per taught course (`/classes` , migration `0019`); offline write queue (progress/quiz/resume flush on reconnect) + per-week Save offline into a version-proof cache (SW v6).
- **Google sign-in (v1.11.0).** OAuth on both auth tabs, onboarding prefill, no backend change (API is provider-agnostic).

Batch of 20 October 2026 (v1.25.0 → v1.26.13)

- **Reader controls (v1.25.0 → v1.26.3).** Focus mode hides the top bar + bottom tabs while reading (store `lib/focus.ts`, persisted per device). Both Focus and zoom are **retained when switching notes**; the layout only hides chrome on reader routes, so neither ever leaks onto dashboard/studio/etc. The floating bottom-right pill is the only place controls live and renders **only on note readers**: Focus on every device; zoom-in only 100–150% on desktop, targeting `#unify-content` so header/nav keep normal size; first zoom-in switches Focus on, the % button resets zoom and exits Focus; leaving a reader clears the applied zoom.
- **Dropdowns (v1.25.0/v1.26.0).** `Drop` overlay component (floating panel, outside-click/Escape, flip-up near viewport) replaced native selects app-wide with shared option spacing; the read-aloud voice picker was un-squeezed from its flex row.
- **Math rendering (v1.25.0/v1.26.0/v1.26.1).** MathJax typeset race fixed (awaits `startup.promise` + polls `typesetPromise`). Studio review and NoteBuilder previews render **real formulas** through `<Formula>` again (plain-text preview output was the "code-like" complaint). Offline/CDN-blocked fallback hardened: spaced AI LaTeX (`\frac {a}{b}` , `\sqrt {x}`) is normalized before glyphing so roots and fractions always show their signs (`√(x)` , `a/b`) — previously they degraded to literal `frac ab` / `\sqrt x` .
- **Never-raw math + offline previews (v1.26.4) — RESOLVED, closed.** The "formula section still shows `\frac` " report is fixed and verified shipped: `typesetMath` swallowed every failure (a rejected typeset on malformed TeX, a hung MathJax font fetch, a CDN timeout) and `MathText` had no fallback at all, so raw source sat on screen forever. Now `typesetMath` returns a boolean with a hard 6s deadline + one clear-and-retry; `Formula` renders glyph math from first paint and only reveals an off-screen MathJax twin after a real typeset (or a **saved render** from `localStorage unify.fx.v1` , capped at 80 entries) — raw TeX is never displayed online or offline; `MathText` swaps failed `\(...\)` segments for glyphs; the NoteBuilder equation field gained a persistent live `RENDERED` preview box (it previously showed only the builder button until you opened the visual builder). Offline preview is no longer hidden or empty — it shows the saved typeset render, or readable glyphs when nothing is saved yet.
- **Email (v1.25.0).** Profile shows the email-verify chip with a Resend action (`email_confirmed_at`); new-note + announcement emails send from Render's `EMAIL_FROM` via the Brevo API (see SETUP.md — the from-address must be a verified Brevo sender or every send fails `brevo 400`).

- **Email delivery + Drop spacing (v1.26.12).** Announcements fired up to 50 concurrent Brevo calls, so the free-tier rate limit rejected bursts (429) and every rejection body was thrown away - the admin saw "sent to N users" because only the in-app bell was counted. Sends now batch 5 at a time with 429/5xx retry (honouring `Retry-After`), the recipient walk no longer stops early on emailless rows, the announce response reports how many confirmed inboxes were queued, and Admin > Announce gained an **Email check** that sends one real email and prints Brevo's exact status + body (with `BREVO_API_KEY` / `EMAIL_FROM` hints). `Drop` now applies form-field styles to the trigger instead of the wrapper, so the course picker matches the Week input in height, and the panel widens to show full course titles instead of ragged truncation.
- **Bare LaTeX renders everywhere (v1.26.13).** The reported "raw `\frac`" came from paragraph math: `MathText` only rendered spans wrapped in `\(...\)`, while AI output (and stored notes) often leaves TeX undelimited - so `\frac{v^2}{2g}` sat as source in paragraphs, titles, bullets and quiz text. The segmenter now finds bare commands (`\frac`, `\sqrt`, `\sum` with limits, Greek symbols...) outside code spans, KaTeX renders them instantly, and anything KaTeX refuses is wrapped and handed to MathJax. `$$...$$` is supported too; sentence punctuation and surrounding prose are left alone. 12-case node sweep covered fractions, roots, sums, AI-spaced `\frac {a}`, display math, `<code>` exclusion, prose boundaries and the MathJax hand-off.
- **Avatars (v1.25.0/v1.26.0).** Any-size upload: input is downscaled on-device to a 256px JPEG; backend accepts png/jpe?g/webp/gif/avif/bmp data URLs and picks the real format from magic bytes (declared MIME never trusted); `?v=` cache-bust + `unify:avatar` event keep the header chip live. `/me` backfills Google/Apple SSO profile pictures (hourly throttle). Server rejections now surface verbatim in the profile UI.
- **Save PDF (v1.26.6).** The XP-gated Save PDF button is `window.print()` of the rendered reader — documented in apps/web/README (it had no docs entry). Print cleanup: the duplicated `@media print` block was merged into one, the ZoomBar pill and InstallBar are now `screen-only` (they used to print as empty floating boxes), and `#unify-content` zoom is forced back to 100% in print so a zoomed session never clips the exported page. Because the reader is never-raw since v1.26.4, PDFs can only contain typeset math or glyphs — never TeX source.
- **Delivery speed (v1.26.0).** Route-level code-splitting (studio/admin/browse/share/classes/onboarding/notifications/explore are `React.lazy`): main bundle 350→190 KB raw (89→52 KB gzip). jsDelivr preconnect for MathJax. Assets already immutable-cached on Vercel; SW documents stay network-first.
- **Docs (v1.26.1+).** README, apps/web/README, SETUP (BREVO/EMAIL_FROM env + no-mail troubleshooting), UNIFY_MATHJAX_RULE (React implementation + canonical LaTeX spacing), product doc v1.26.7 and this report refreshed; versioning rule (bump `APP_VERSION` every push) recorded.

- **Direct typesetting restored (v1.26.8).** Per user request the v1.26.4 glyph-first twin was removed: `Formula` typesets directly again (raw TeX in, MathJax rewrites in place — the renderer that always worked). Kept from v1.26.4: the boolean `typesetMath` with a 6s deadline + one retry (failures fall back to glyphs instead of leaving raw code on screen forever), saved offline renders (`unify.fx.v1`), the `MathText` failure swap and the live RENDERED preview in the equation field.
- **KaTeX restored for formula boxes (v1.26.9).** Per user request the renderer from before MathJax is back: `Formula` now renders with KaTeX (npm-bundled, served from our own origin — no CDN, no typeset race, works offline by construction, lazy-loaded code-split chunk). TeX KaTeX can't parse falls back to the glyph text, never raw commands. Paragraph inline math (`MathText`) stays on the MathJax pipeline unchanged.
- **Full KaTeX switch, MathJax removed (v1.26.10).** User reported glyphs where formulas should render after v1.26.9. Two causes: (1) KaTeX was lazy-loaded as a separate chunk — on flaky/offline data `import('katex')` rejected and everything fell to glyphs; (2) paragraph inline math (`MathText`) was still on the MathJax CDN and failed exactly like the original BUG-002. Fix: KaTeX is now statically imported (rides inside the main bundle the app already fetches — one request, offline by construction) and `MathText` renders each `\(...\)` / `\[...\]` segment through KaTeX synchronously. `src/lib/mathjax.ts` deleted, jsdelivr preconnect removed, `escHtml` moved into `Formula.tsx`. MathJax is no longer used anywhere in the app.
- **KaTeX + MathJax hand in hand, no fallbacks (v1.26.11).** Per user: MathJax must still render complex formulas and previews must be exact. One two-stage pipeline everywhere: KaTeX (static bundle) first, MathJax (on-demand, SW-cached) for whatever KaTeX rejects; `Formula` and `MathText` share it, and the equation editor's RENDERED preview runs the same component as the reader. The glyph engine was deleted outright — only a double failure leaves source (logged). Also in this push: `Drop` dropdown spacing polished (row rhythm, optical alignment with the trigger, panel offset) and email failures now log Brevo's actual response body + sender/recipient context + an `EMAIL_FROM`-not-set warning (before: bare `brevo 400` with no explanation).

Left + verification log

1. **Render redeploy — VERIFIED 4 Oct 2026.** Migrations `0016`–`0020` applied live; new endpoints (aliases search, lecture paths, timetable, roster, error log, support config) confirmed working in production.
2. **Google sign-in live test — VERIFIED 4 Oct 2026.** Real Google account in through onboarding to dashboard after the console enablement.
3. **Lecturer walkthrough — VERIFIED 4 Oct 2026.** Teaching account: courses assigned in admin, slots set, roster visible, student-side class-times card correct.

4. **Phone pass — VERIFIED 4 Oct 2026.** Student + contributor sides on real phones.
5. **Persistence retest — VERIFIED 4 Oct 2026.** Resume (with lecture), streaks, and offline queue flush confirmed on device.
6. **Proposed, not built:** per-university allowed-email-domains policy (the actual gate for multi-school expansion — works with both email and Google paths), and a link-Google-to-existing-account action (same address via both methods can mint two identities).
7. **Open product decisions (need a ruling, not testing):** per-lecture EOQ banks vs the single week bank after the final lecture; removing Explore into the dashboard per BUG-012's ideal vs keeping it as the add-courses path.

Brand direction: why the generated kit can't hold as law

The `docs/Unify-Brand-Kit/` (posted 2 October 2026) was single-pass AI output (Claude): no user testing behind it, no ratification, and it contradicts decisions that were already tested and shipped. A generated draft is an input to a brand decision — it is not the decision. What follows is the recorded reasoning so the next reader doesn't re-litigate it from a chatbot's authority.

1. The kit contradicts the product's own retention thesis. The master PRD's core loop is Duolingo mechanics (streaks, XP, badges, tactile progression) applied to university study. The kit's headline rule — "it should never look like a game" — outlaws the very psychology the product is built on. Tested reality agrees with the PRD: residents and students shown the Duolingo-style revamp (Classic) preferred it. Playfair editorial (Story) was added as a second theme to keep the serif direction alive, not to replace a validated winner.

2. The kit contradicts tested user requests. It mandates one-column course cards; real testing (BUG-012) asked for a card grid. It bans Save PDF; the XP-gated PDF is a shipped retention incentive users understand. It bans the mascot from the dashboard; the greeting mascot is part of the daily habit loop. In each case the tested artifact wins over the generated rule — the kit's own social guideline concedes this precedent (the shipped lighter post template "is now the standard" over the earlier generated one).

3. The case for the Duolingo style (Classic, the default). Duolingo (~100M+ MAU), Quizlet and Khan Academy converge on the same findings for young mass-market learners: chunky 44px-grade touch targets (cheap Androids, touch imprecision), high-contrast CTA colour (sunlight legibility), rounded friendly type (Nunito stays legible at 12px where thin-serif Playfair strokes break on low-end screens), and celebration moments (XP bursts, streaks) that create the return habit the PRD demands. Serif-editorial is the right voice for lecturers, certificates and campaigns — not for a first-year student revising on a bus.

4. Why the theme switch stays. Two validated tastes exist and they split by role and age: students motivate on Classic; lecturers, collaborators and formal surfaces read premium on

Story. Forcing one face loses one audience. Both themes persist, per-device, with usage measured — a future default follows data, not taste. Proposed instrumentation: log the active design with existing analytics events so the split is evidence, not anecdote.

5. The Unify ecosystem brand (adopted). Classic ("Play") and Story ("Editorial") are co-equal official themes, not a main and a fallback. From the kit we adopt what is genuinely good regardless of theme: plain-spoken button labels, zero hype register, visible focus states (accessibility fix queued), the social-post template structure, and Story's token set for the editorial theme. From the kit we reject: the Playfair-only rule, the gradient ban where gradients carry meaning (progress, earned states), the PDF ban, the mascot confinement, and the one-column-cards rule. Future brand proposals arrive as diffs against this section with test evidence attached — never as generated doctrine.
